MTH636 – Game Theory

Lecture 11: Computing Max-Min Strategies graphically

1 Max-Min Strategies in Mixed Games

For a finite two-player zero-sum game, we know the value V exists in mixed strategies:

$$V = \max_{\sigma_1 \in \Sigma_1} \min_{s_2 \in S_2} U(\sigma_1, s_2) = \min_{\sigma_2 \in \Sigma_2} \max_{s_1 \in S_1} U(s_1, \sigma_2)$$

Definition: Optimal Strategies

- For Player 1, a **max-min strategy** is a mixed strategy σ_1^* such that:

$$U(\sigma_1^*, \sigma_2) \geq V \quad \forall \sigma_2 \in \Sigma_2$$

- For Player 2, a **min-max strategy** is a mixed strategy σ_2^* such that:

$$U(\sigma_1, \sigma_2^*) \leq V \quad \forall \sigma_1 \in \Sigma_1$$

2 The Graphical Method (2×2 Games)

When Player 1 has exactly two pure strategies, we can find the max-min strategy by graphing the expected payoffs.

Example: A 2×2 Game Example

$P_1 \setminus P_2$	L (y)	R $(1-y)$
T (x)	5	0
B $(1-x)$	3	4

Let Player 1 play T with probability x and B with probability $1-x$. Player 1's expected payoff against P2's pure strategies:

$$U(x, L) = 5x + 3(1-x) = 2x + 3$$
$$U(x, R) = 0x + 4(1-x) = 4 - 4x$$

We graph these two lines for $x \in [0, 1]$. For any chosen x, Player 2 will respond by choosing the column that yields the **lower** payoff for Player 1. Thus, Player 1's guaranteed payoff for a given x is the **lower envelope** of the two lines:

$$\min\{2x + 3, 4 - 4x\}$$

Player 1 wants to maximize this minimum (find the peak of the lower envelope). This peak occurs where the two lines intersect:

$$2x + 3 = 4 - 4x \implies 6x = 1 \implies x^* = \frac{1}{6}$$

The unique max-min strategy for Player 1 is $x^* = 1/6$. The **Value** of the game is $V = 2(1/6) + 3 = 10/3 = 3\frac{1}{3}$.

By symmetry, to find Player 2's min-max strategy y (probability of L), we graph P1's payoffs against P1's pure strategies:

$$U(T, y) = 5y + 0(1 - y) = 5y$$
$$U(B, y) = 3y + 4(1 - y) = 4 - y$$

Player 2 wants to minimize the **upper envelope** (because P1 will maximize). The intersection is $5y = 4 - y \implies 6y = 4 \implies y^* = 2/3$.

3 The Graphical Method ($2 \times n$ Games)

The graphical method also works if one player has 2 strategies and the other has n.

Example: A 2×3 Game Example

$P_1 \setminus P_2$	L	M	R
T (x)	2	5	-1
B $(1 - x)$	1	-2	5

Player 1's expected payoff against P2's three pure strategies:

$$U(x, L) = 2x + 1(1 - x) = x + 1$$
$$U(x, M) = 5x - 2(1 - x) = 7x - 2$$
$$U(x, R) = -x + 5(1 - x) = 5 - 6x$$

Graphing these three lines for $x \in [0, 1]$, we look at the **lower envelope**. The highest point on the lower envelope occurs at the intersection of the lines for L and R. (The line for M is strictly above this intersection point, so it does not restrict Player 1's maximum guaranteed payoff).

$$x + 1 = 5 - 6x \implies 7x = 4 \implies x^* = \frac{4}{7}$$

The unique max-min strategy for Player 1 is $x^* = 4/7$. The **Value** of the game is $V = U(x^*, R) = 5 - 6(4/7) = 35/7 - 24/7 = 11/7$.

3.1 Finding Player 2's Mixed Strategy

Player 2 must construct a mixed strategy $\sigma_2^* = (\sigma_2(L), \sigma_2(M), \sigma_2(R))$ such that Player 1 can never get more than $V = 11/7$. Because the line for pure strategy M is strictly above the value $11/7$ at x^*, Player 2 should never play M (it would give P1 a higher payoff). Thus, $\sigma_2(M) = 0$.

Let $\sigma_2(L) = q$ and $\sigma_2(R) = 1 - q$. Player 1's expected payoff against this mix must be $\leq 11/7$ for any pure strategy P1 plays:

$$\text{If P1 plays T:} \quad 2q + (-1)(1-q) \leq \frac{11}{7} \implies 3q - 1 \leq \frac{11}{7} \implies 3q \leq \frac{18}{7} \implies q \leq \frac{6}{7}$$

$$\text{If P1 plays B:} \quad 1q + 5(1-q) \leq \frac{11}{7} \implies 5 - 4q \leq \frac{11}{7} \implies 4q \geq \frac{24}{7} \implies q \geq \frac{6}{7}$$

Therefore, $q = 6/7$. Player 2's unique min-max strategy is $[6/7\ (\text{L}), 0\ (\text{M}), 1/7\ (\text{R})]$.